

Linking Surface and Deep Ocean Dynamics and Stressors to Deep-Diving Cetacean Distributions using Long-term Passive Acoustic Data



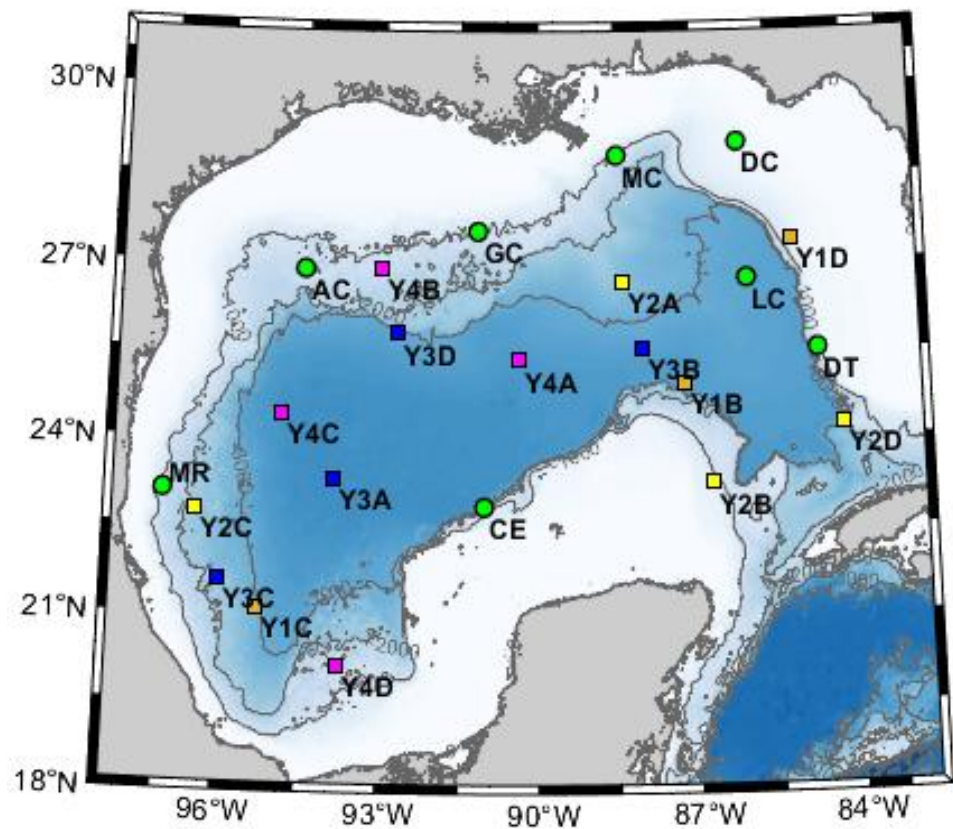
Presented by: Alba Solsona Berga
Kait Frasier, Matthieu Le Hénaff,
Lance Garrison & Melissa Soldevilla



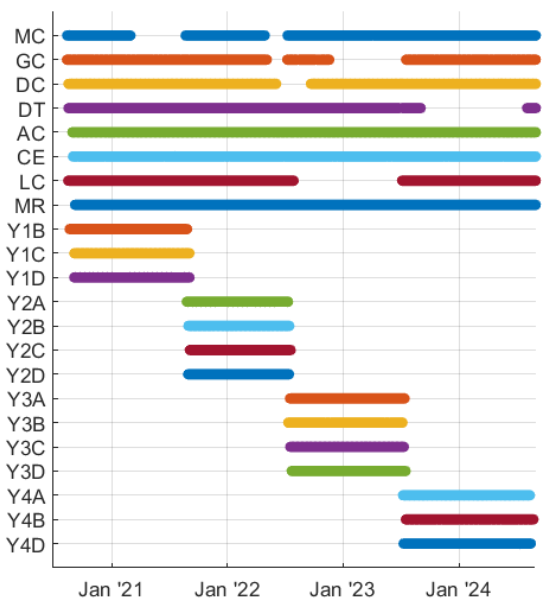
A Gulf wide picture

Passive acoustic monitoring

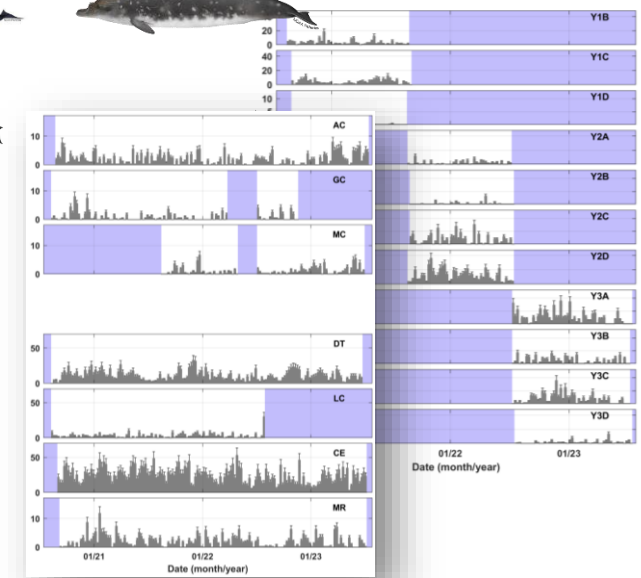
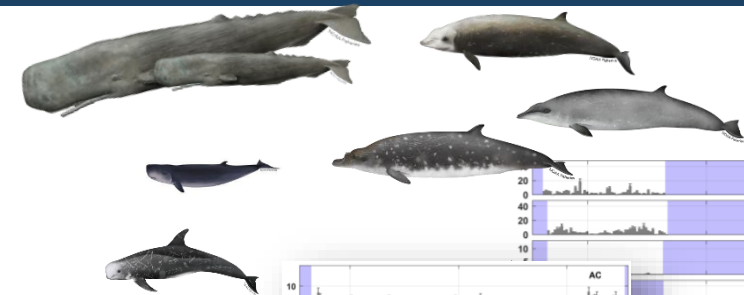
Large-scale



Long-term



A Gulf-wide monitoring network combining long-term and rotating sites to provide temporal and spatial coverage



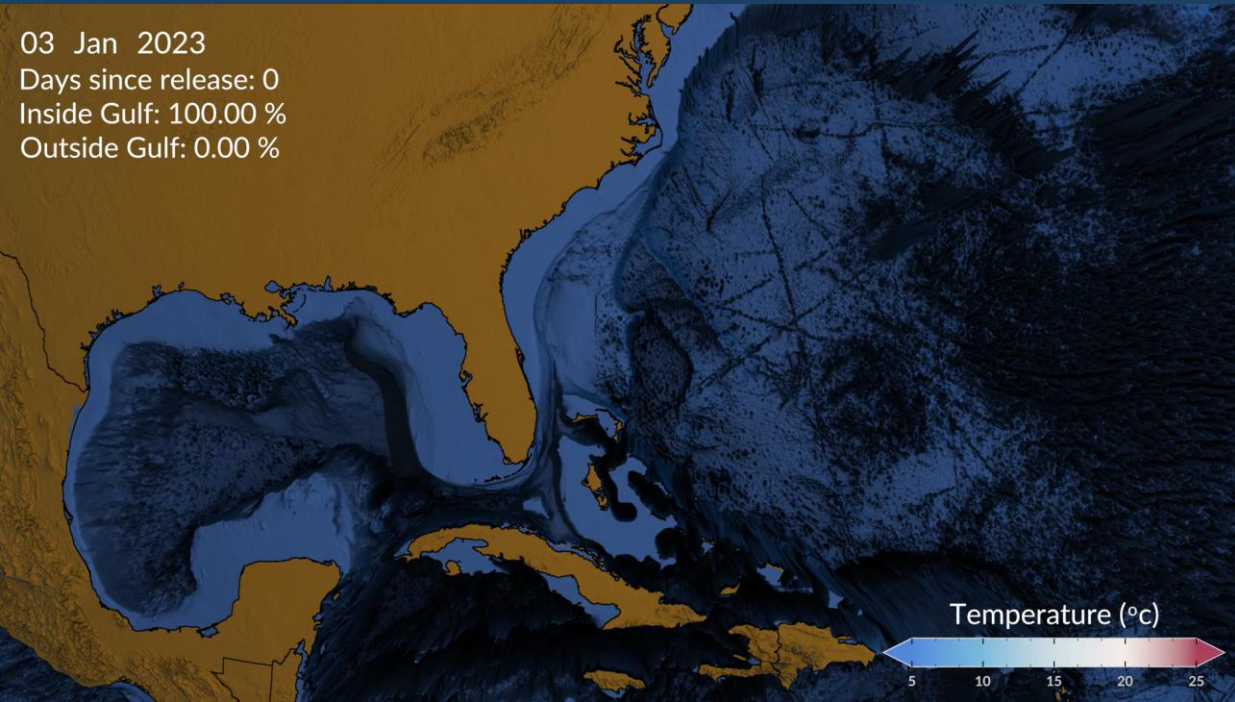
Daily densities of multiple marine mammal species at 21 monitoring sites

7 long-term sites – 2020 to 2024
15 short-term sites – 1 year

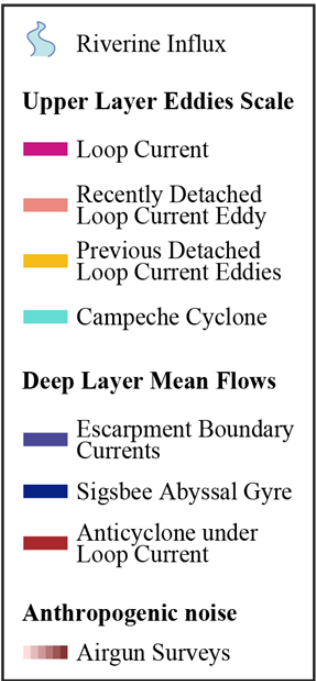
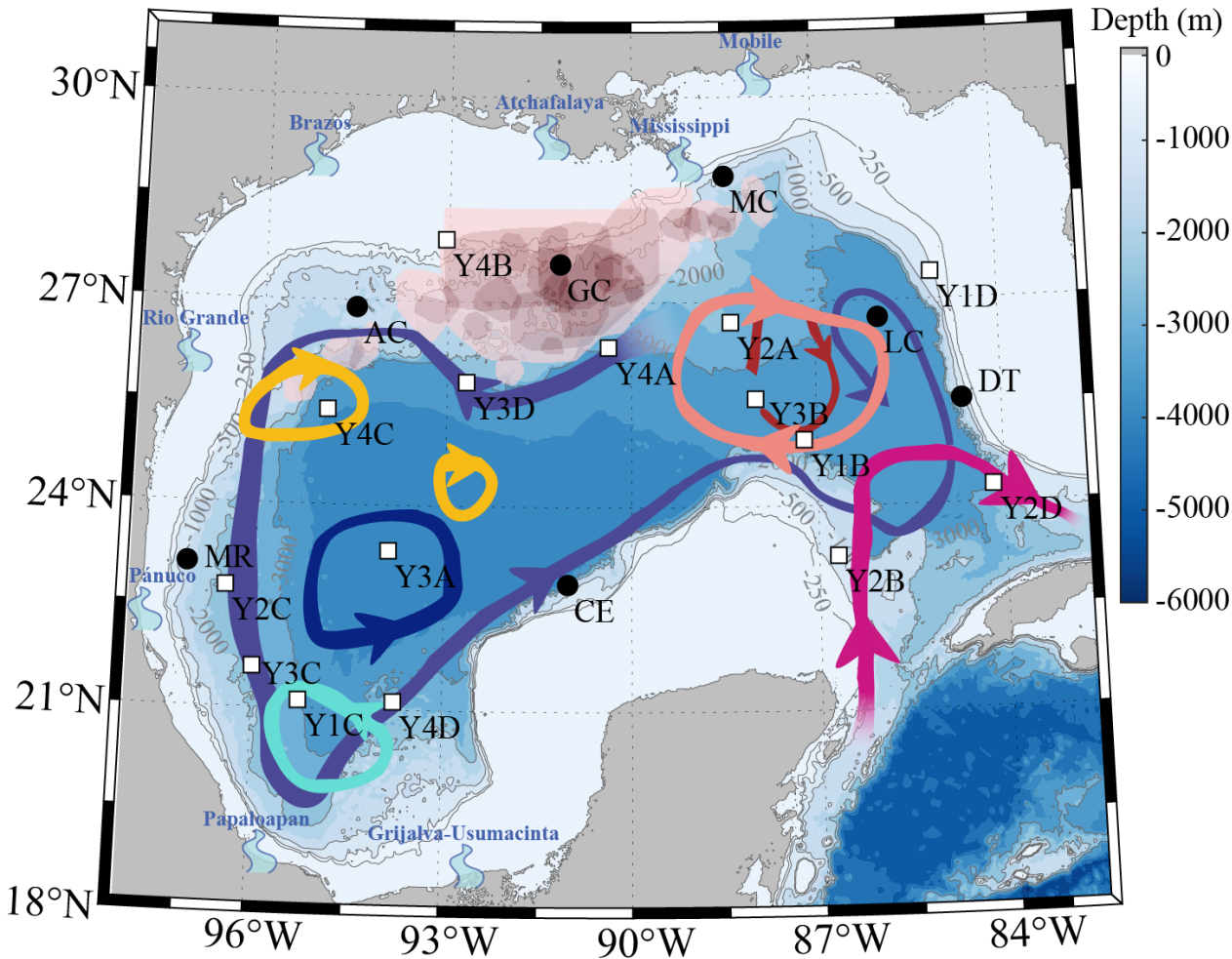
The Gulf Ocean Dynamics

A Complex Ecosystem

03 Jan 2023
Days since release: 0
Inside Gulf: 100.00 %
Outside Gulf: 0.00 %



NASA Scientific Visualization Studio

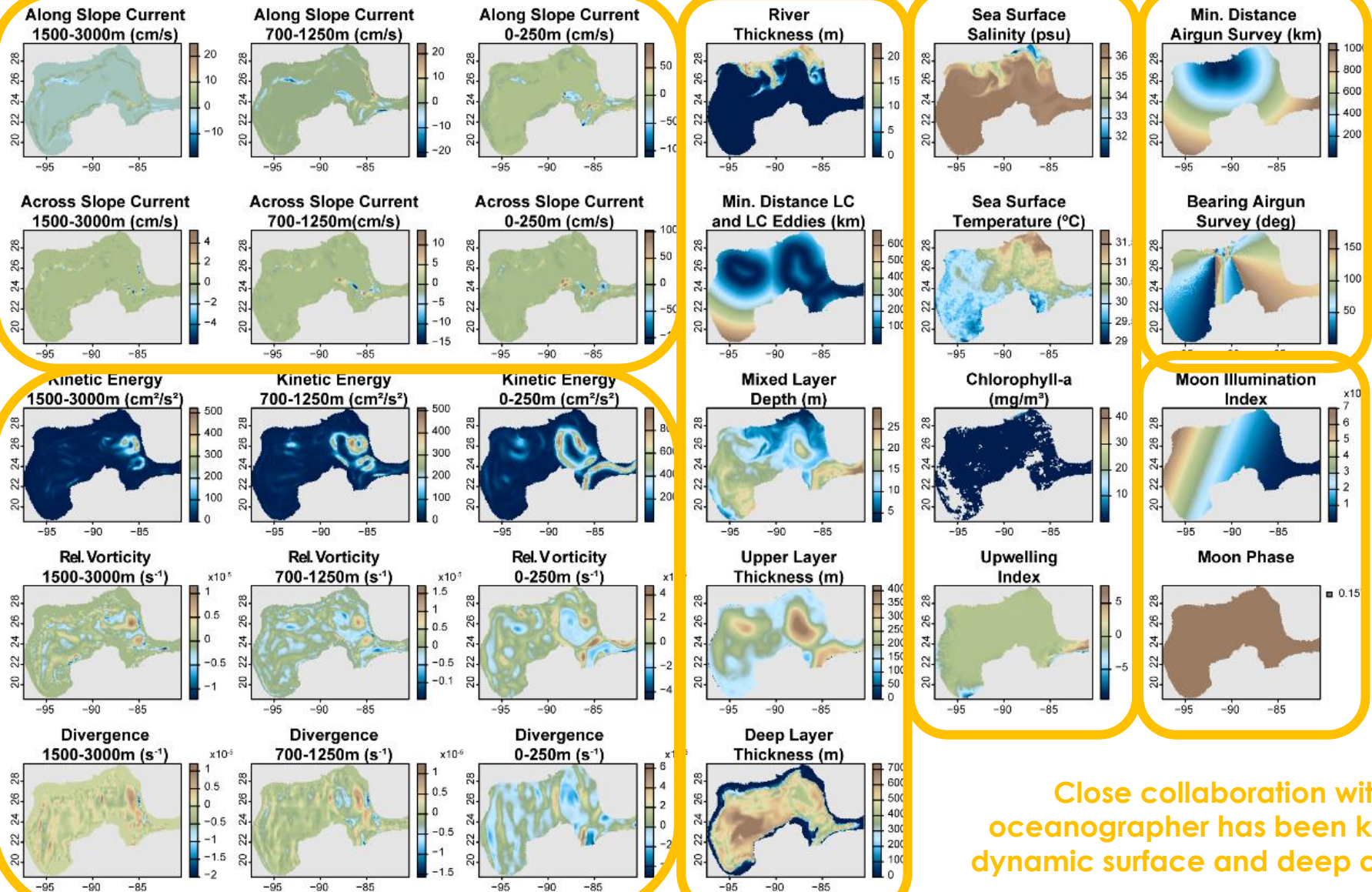


Upper Layer
circulation
(0 to 1000m)

Deep Layer
circulation
(~1000m to
seafloor)

Connectivity
between
upper and
deep
circulation

Environmental Layers:



Close collaboration with physical oceanographer has been key to represent dynamic surface and deep ocean processes

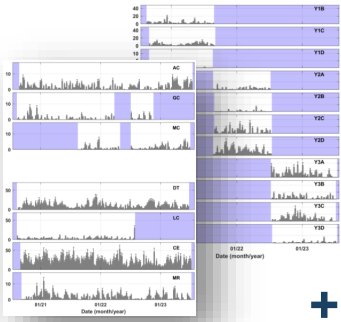
Developing the habitat models

Step 1- Input Data

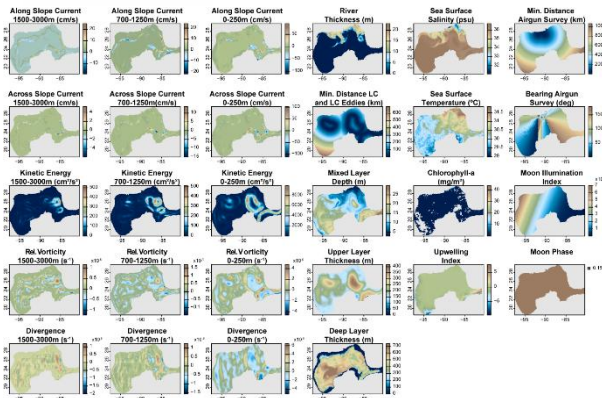
Daily observations across years allow us to resolve responses to dynamic ocean features

Daily:

- Species densities
- Environmental conditions



+



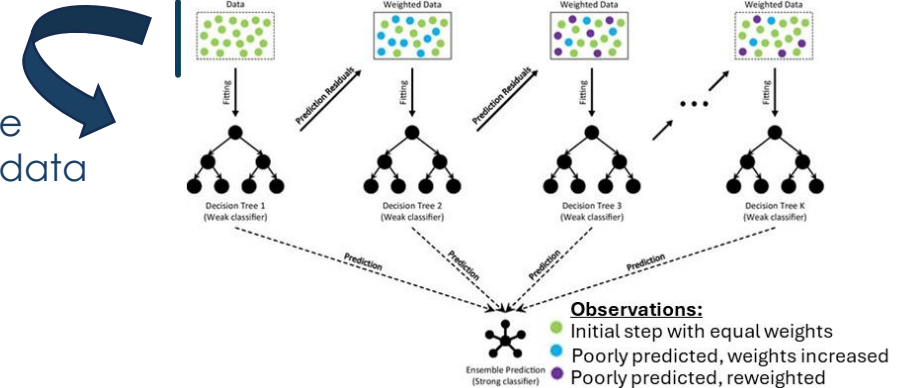
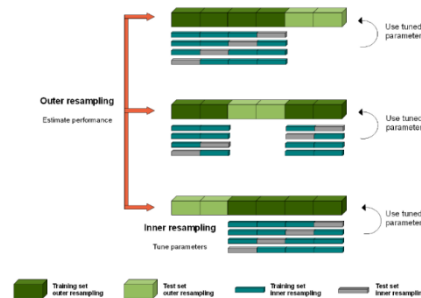
Step 2- Training Machine Learning:

Boosted Regression Trees: XGBoost

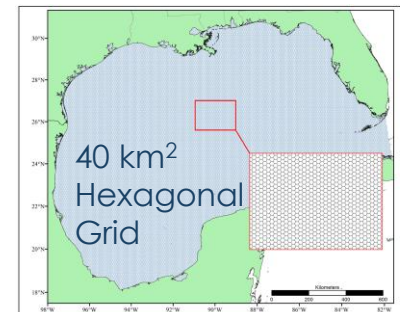
Effective method for capturing complexity in ecological habitats

Resampling Data

Evaluate model's performance and generalization to unseen data

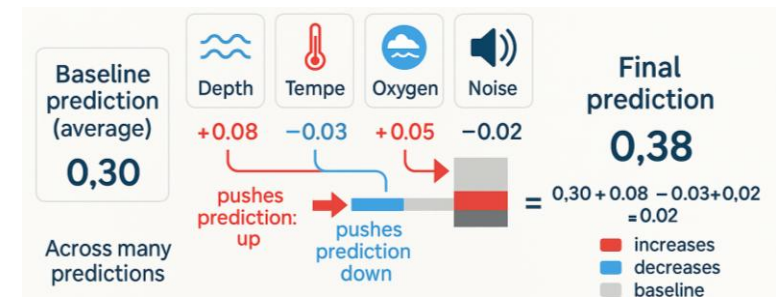


Step 3- Gulf-wide Predictions



Step 4 – Ecological Interpretation

SHAP



Sperm whale

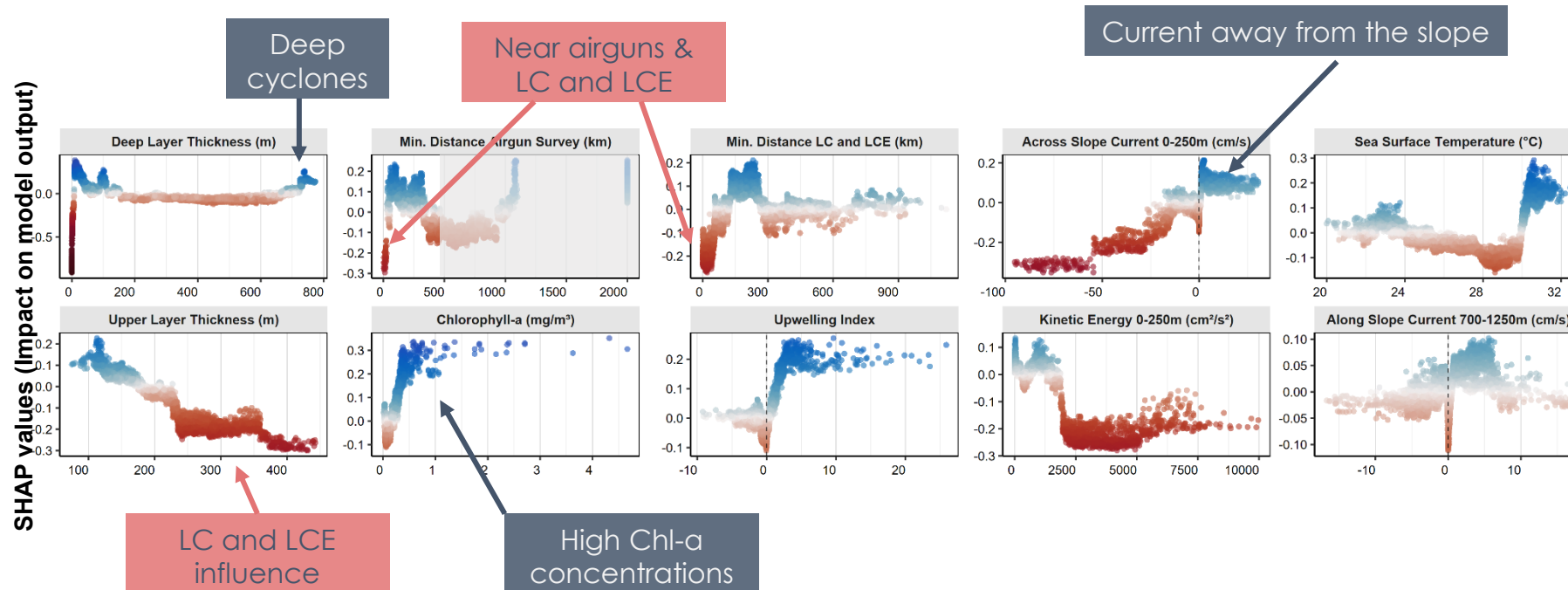


What the model learned from the monitoring sites?

Response patterns

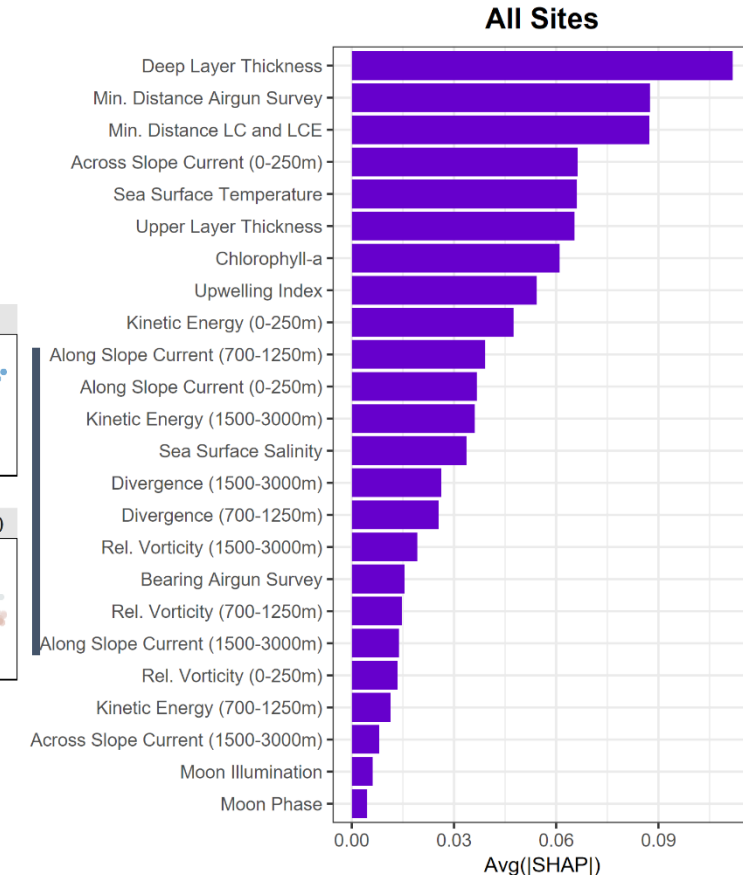
Which predictors values increased or decreased predicted densities

Decreased Increased



Importance

Which predictors contributed most to model predictions



Projected species
densities (ind./1000km²)

Sperm whale



Scaling from local densities to Gulf-wide prediction densities

Daily Gulf-wide predictions

densPm – 13 Aug 2020

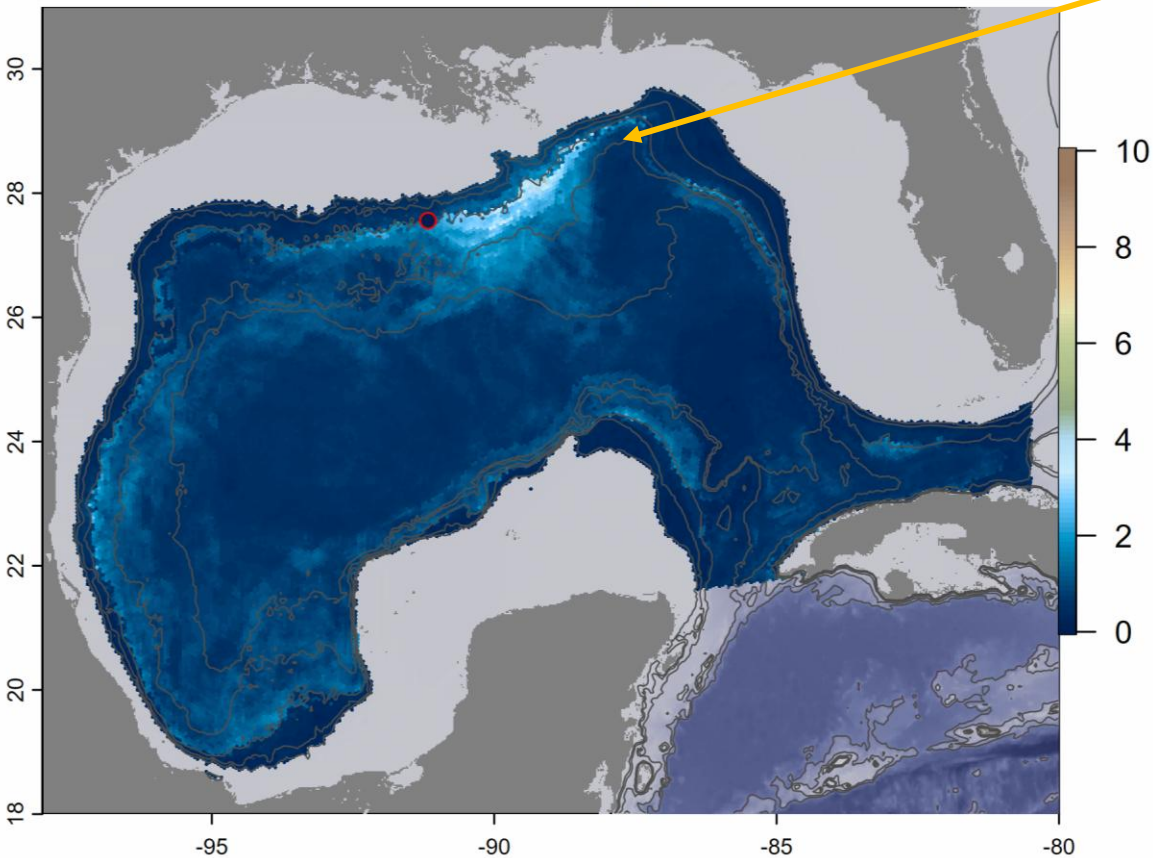
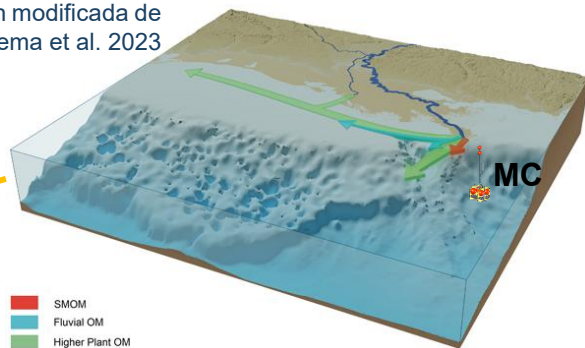
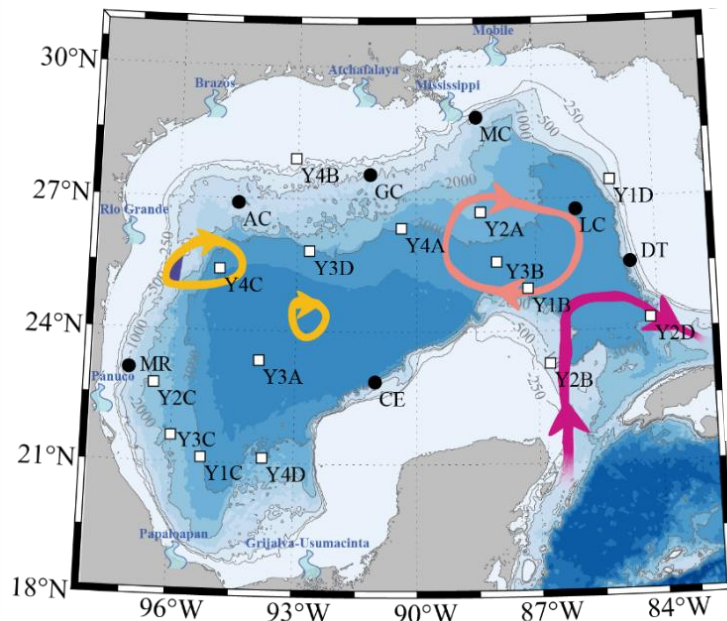
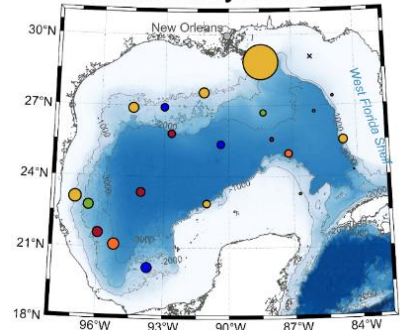


Ilustración modificada de
Yedema et al. 2023



Mean Density 2020-2024



Higher densities in
deep areas near
regions influenced
by river plumes

Lower densities near
the Loop Current and
its shedding eddies

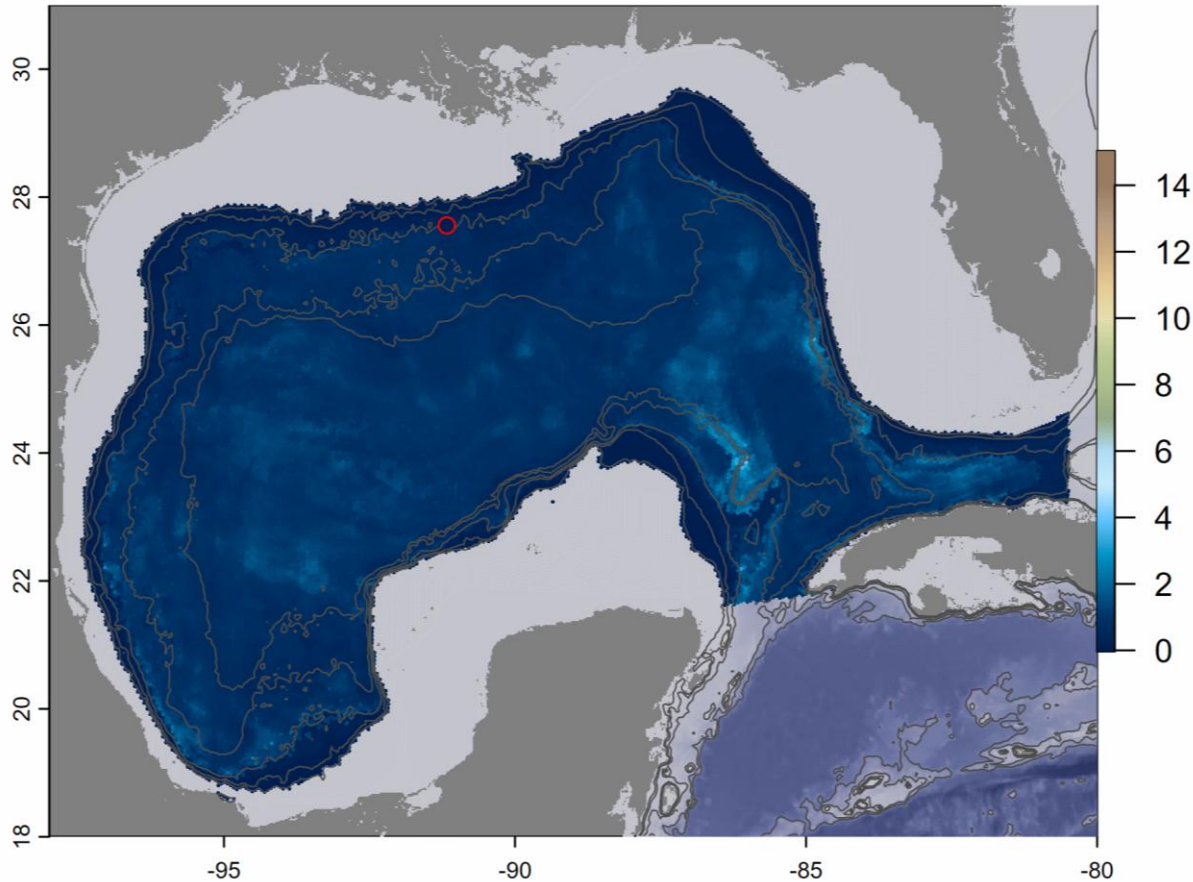
Projected species
densities (ind./1000km²)

Gervais' beaked whale

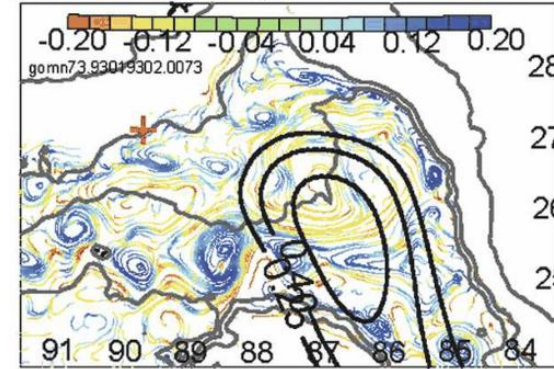


Daily Gulf-wide predictions

densMe – 13 Aug 2020

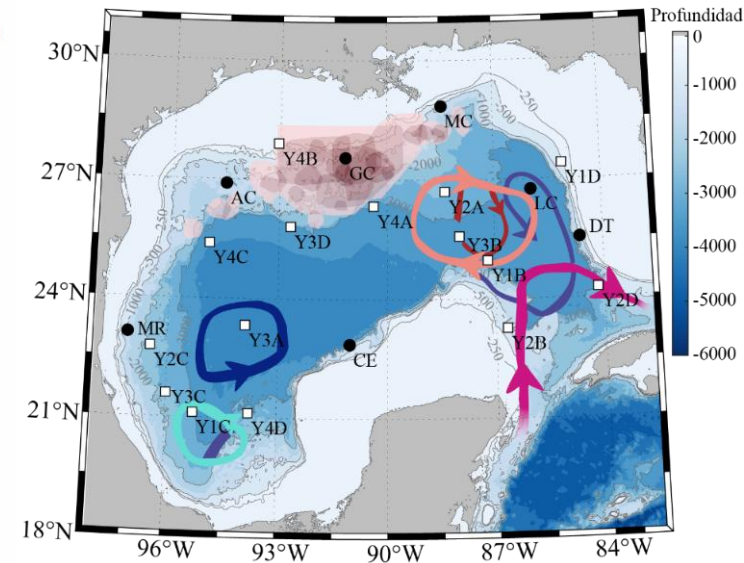


(A) EI & Trj+Vor/f @z=-1550 m 92/12/31



Revolving these features
requires repeated
observations through time

Modified from Oey et al. 2007



Higher densities
in deep cyclonic
eddies.
Coupling of the
upper and deep
circulation

Lower densities
near airgun
surveys

Projected species
densities (ind./1000km²)

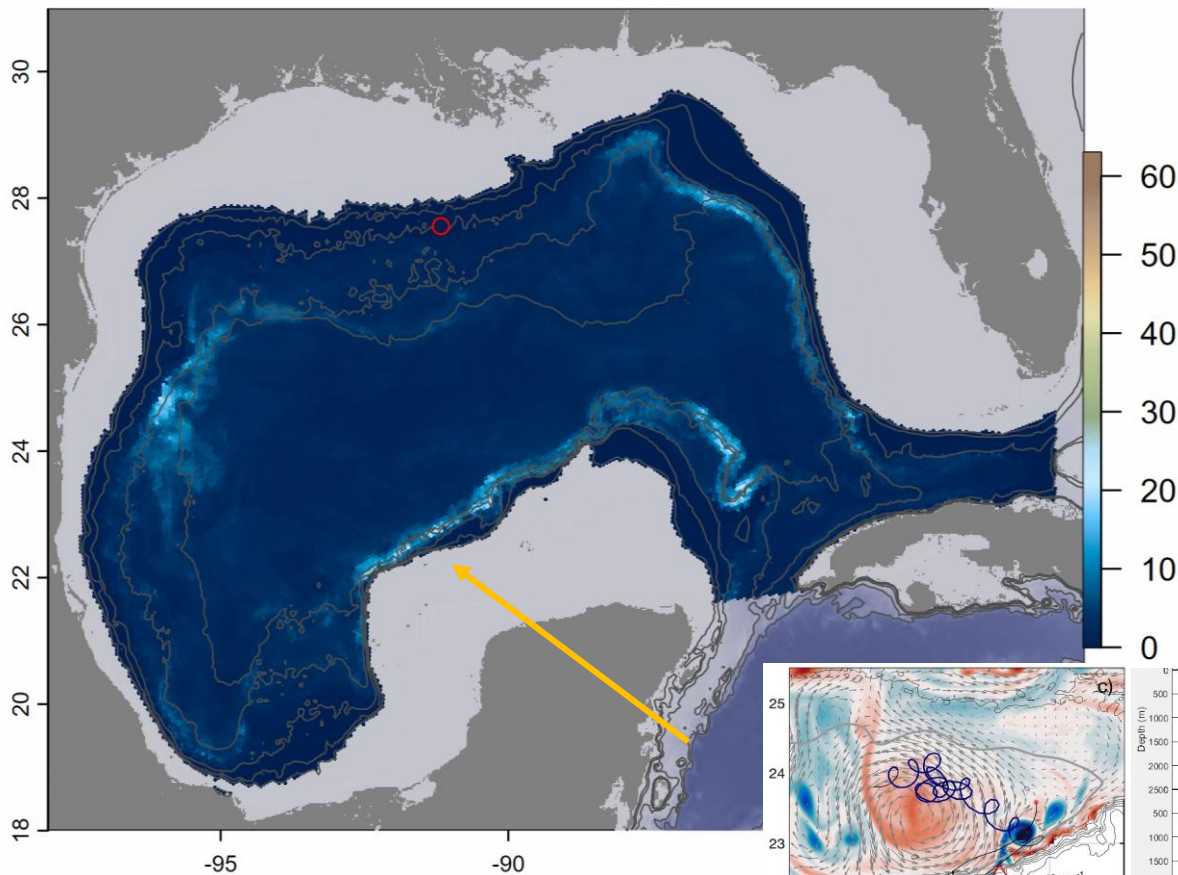
Goose-beaked whale



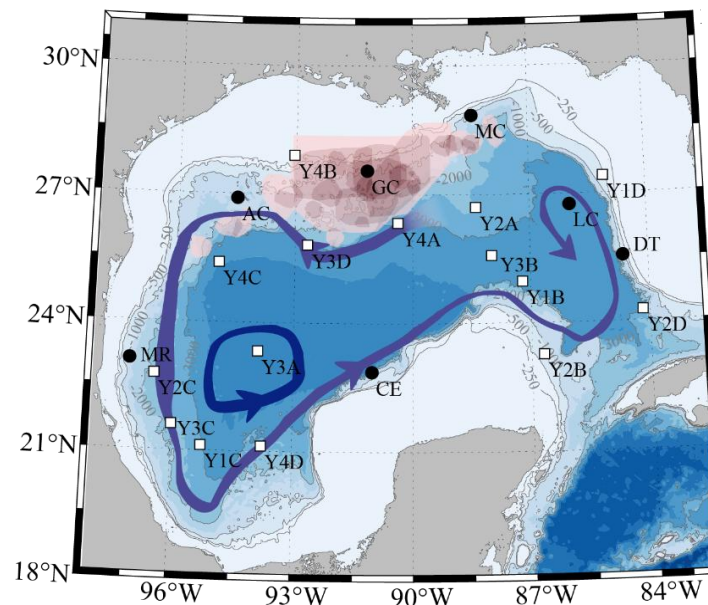
Continuous + spatially distributed observations
let us model species densities across a
dynamic Gulf

Daily Gulf-wide predictions

densZc – 13 Aug 2020

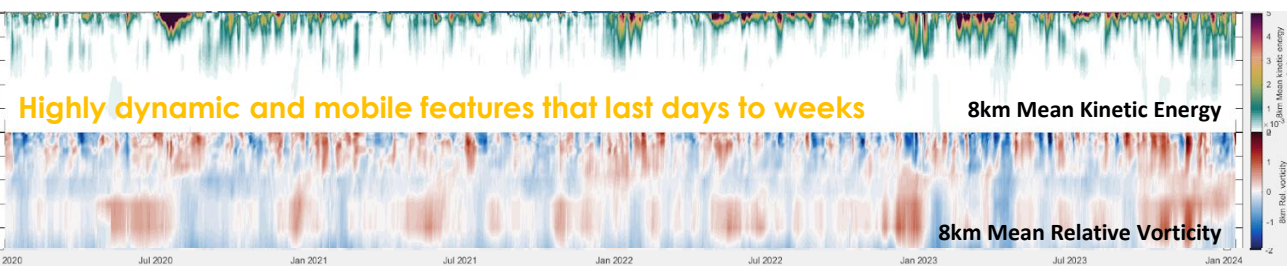


Dominguez-Guadarrama et al. 2025



Higher densities near the
escarpment, where deep
cyclone and deep boundary
currents may create more
stable mixing favoring prey
fields

Lower densities
near airgun
surveys



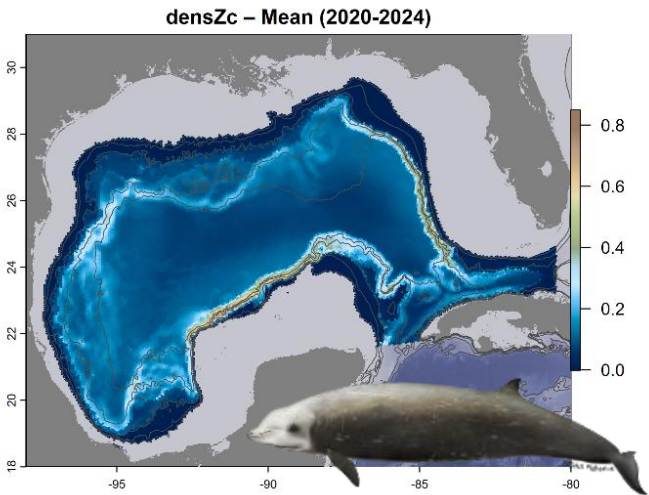
Projected species densities (ind./40km²)

Species Distribution Maps

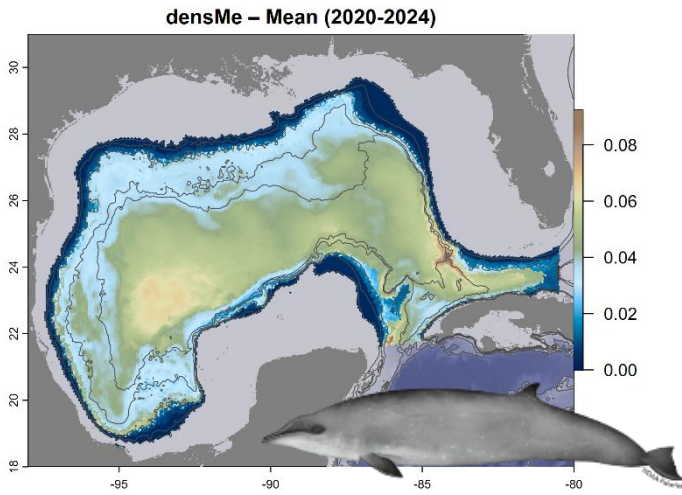
Solsona-Berga et al. in prep

Overall Mean

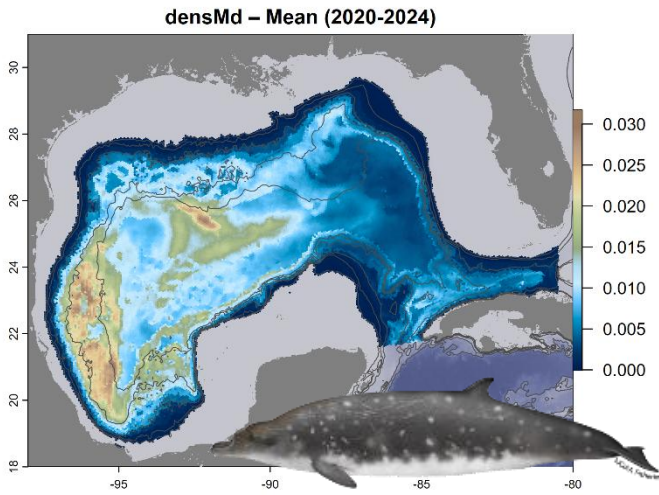
Goose beaked whale



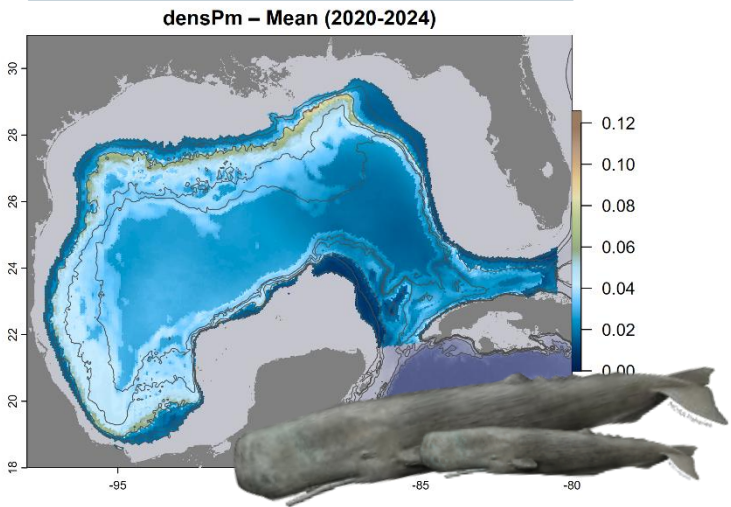
Gervais' beaked whale



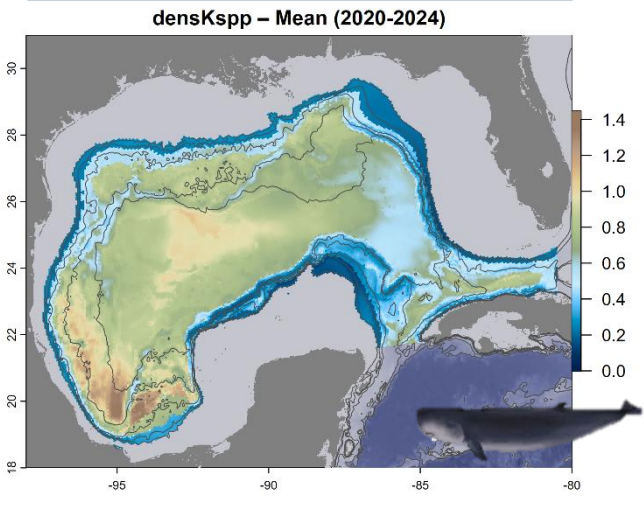
Blainville's beaked whale



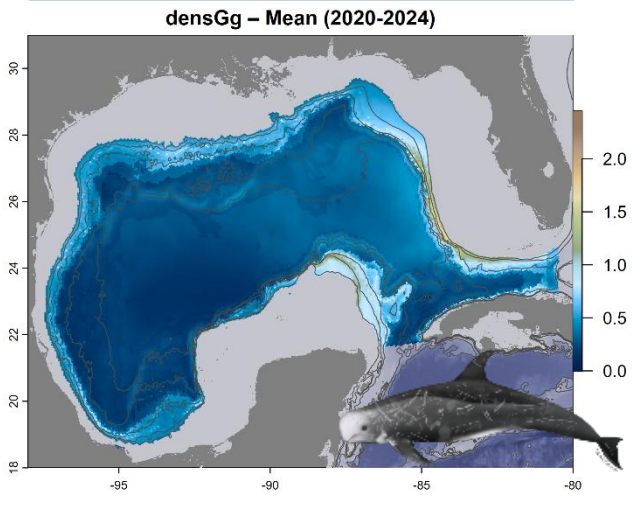
Sperm whale



Kogia spp.

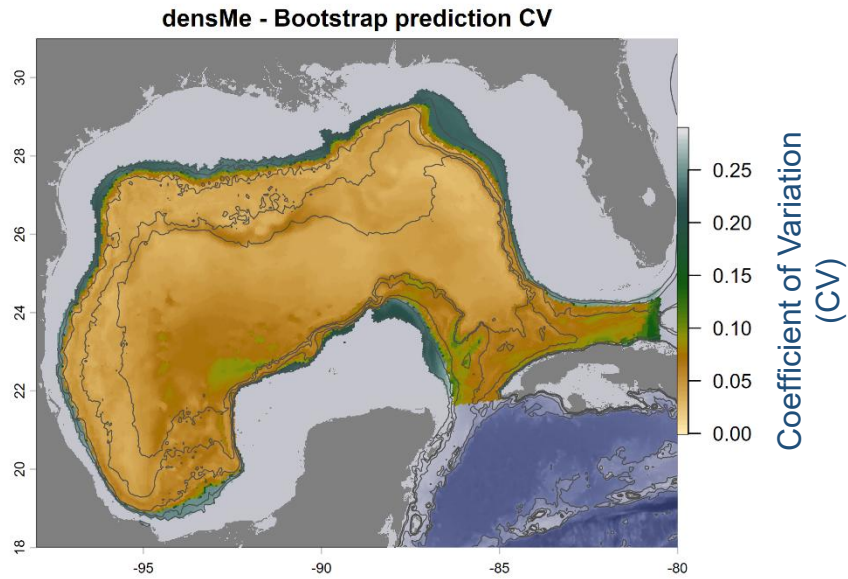


Risso's dolphin



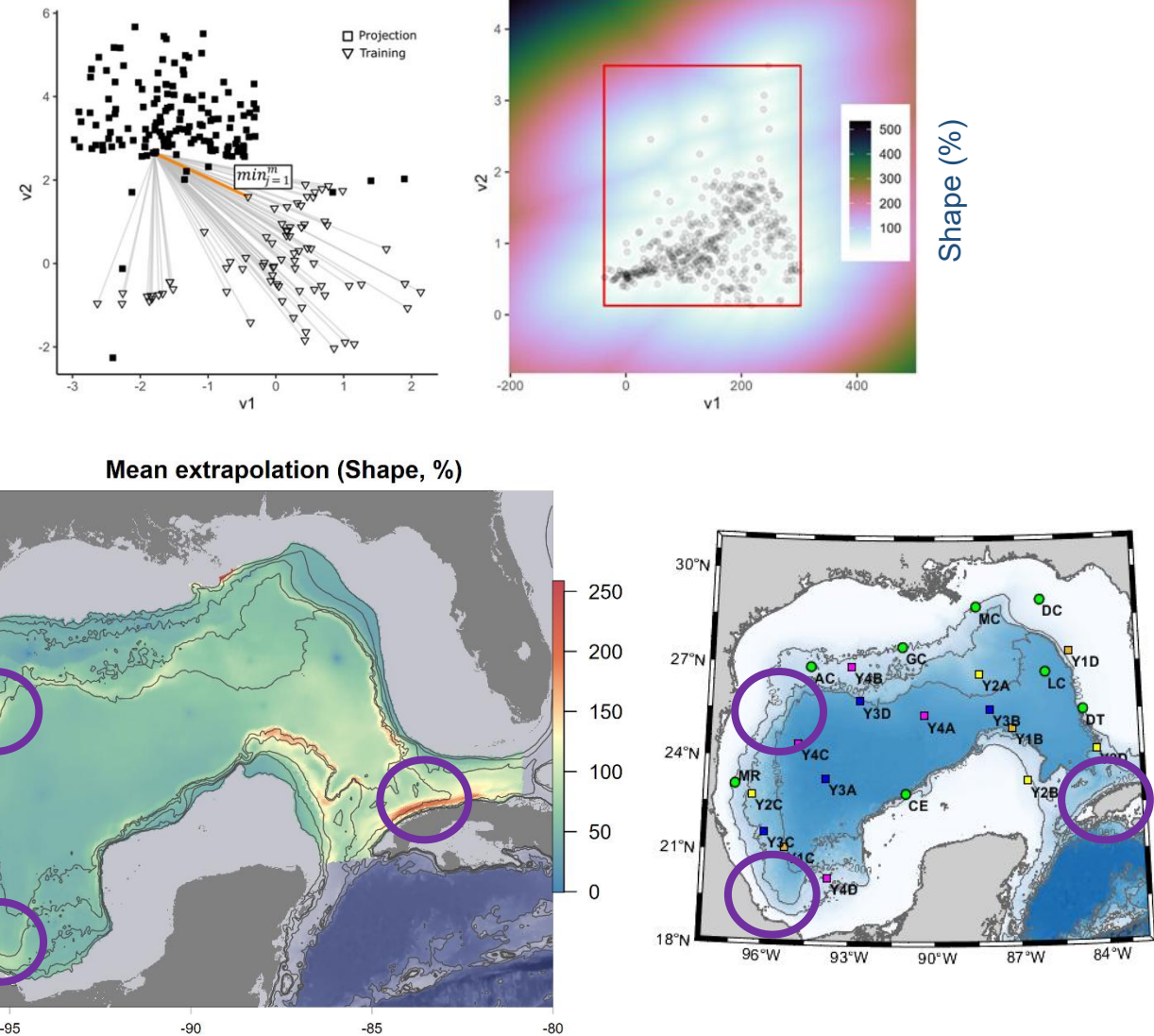
Relative uncertainty

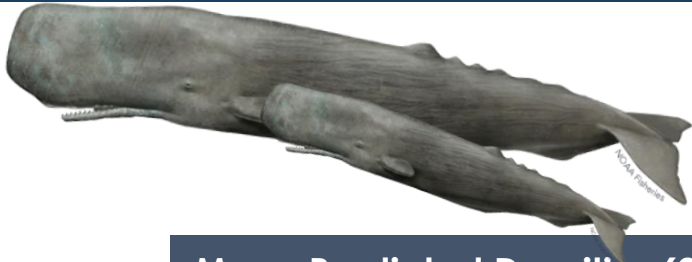
Variation in predicted density across
resampled training datasets
(Bootstrap: 50 iterations, 50% subset)



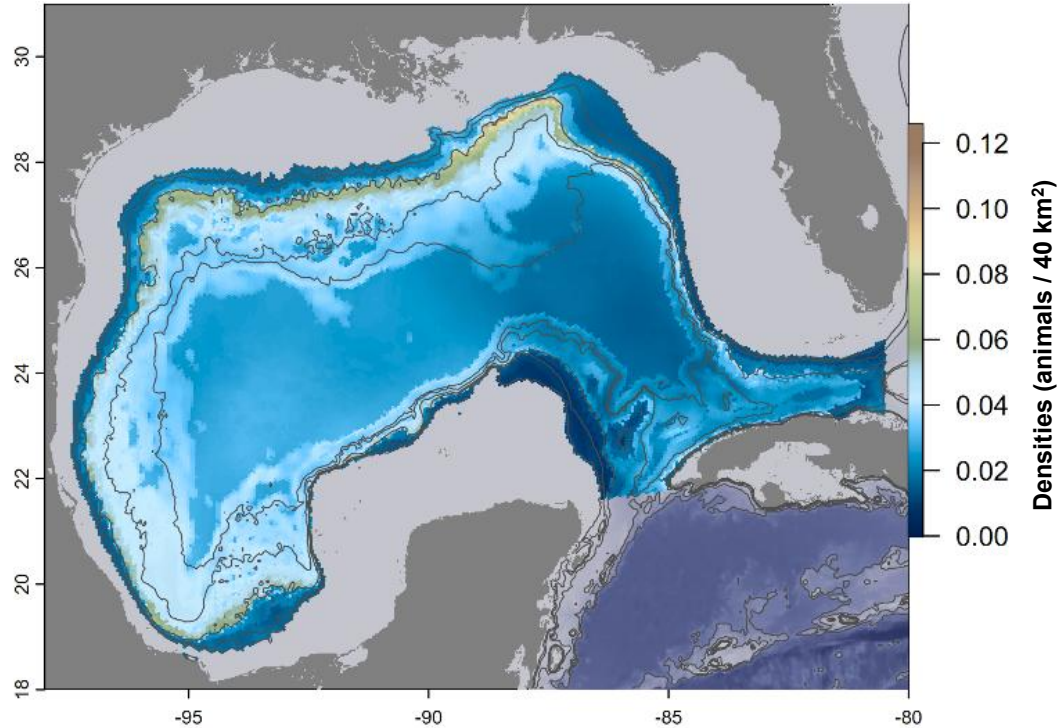
How far can we extrapolate our models?

Velazco et al. 2023



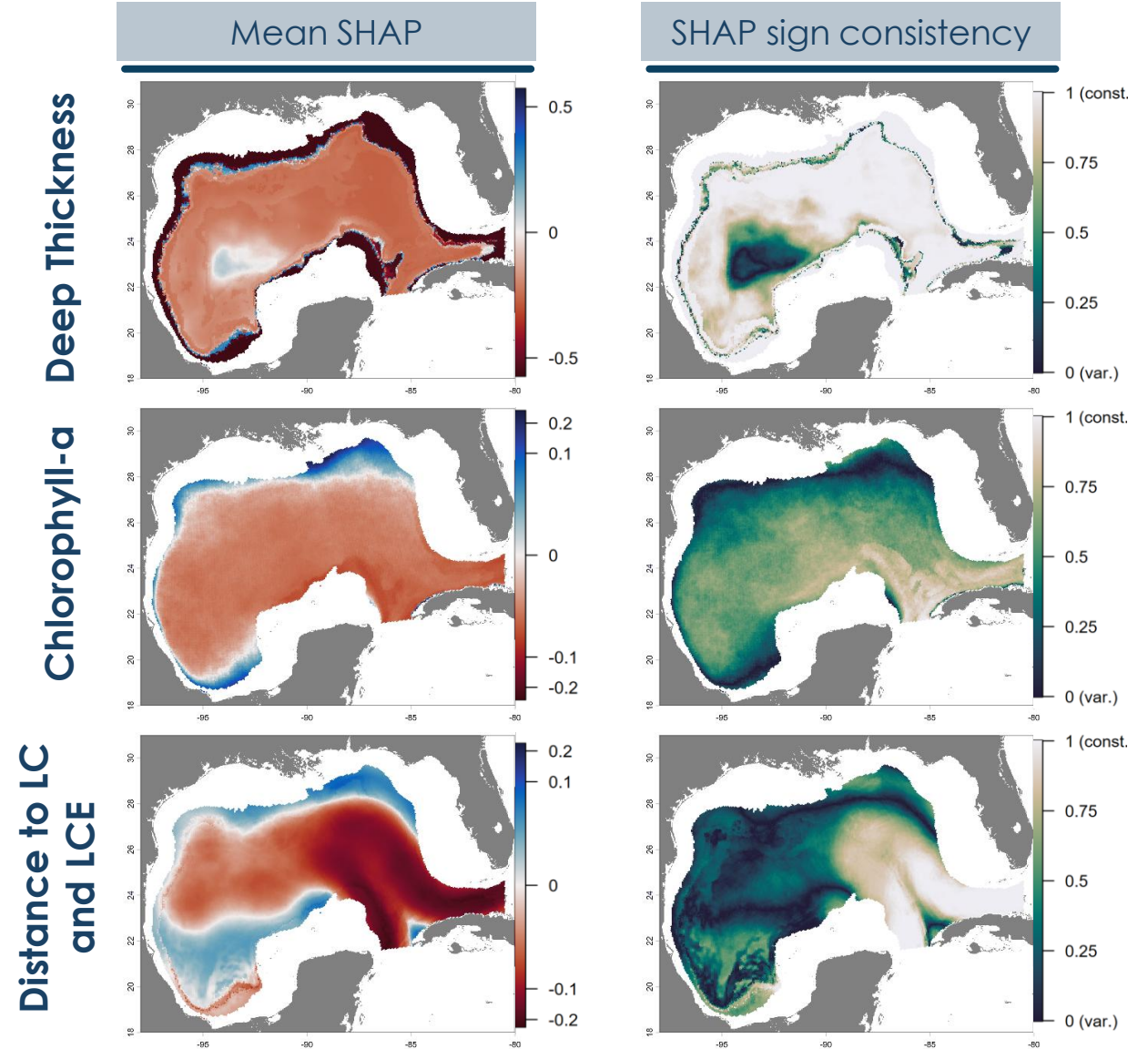


Mean Predicted Densities (2020-2024)



Main predictors linked to higher or lower predicted densities and how consistently that pattern held over four years across the Gulf

Mapping consistent environmental patterns in density predictions



Towards a comprehensive understanding of species distribution and abundance

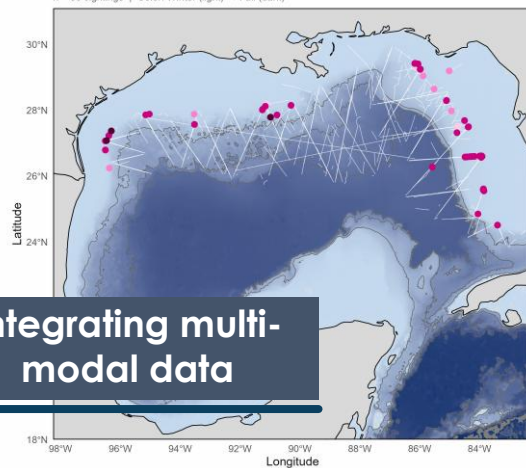
What we are currently doing:

Understanding natural drivers of oceanic species distributions



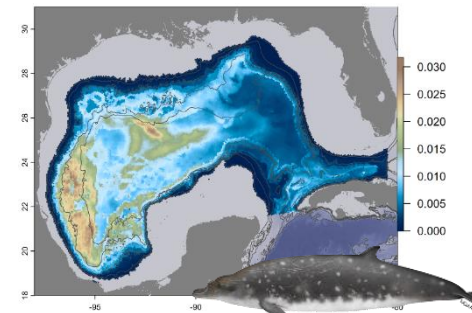
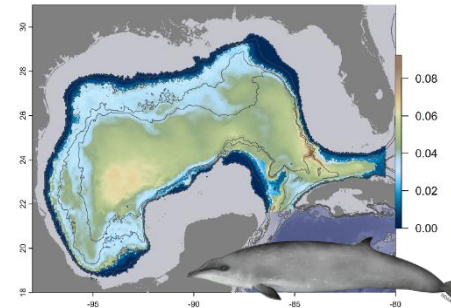
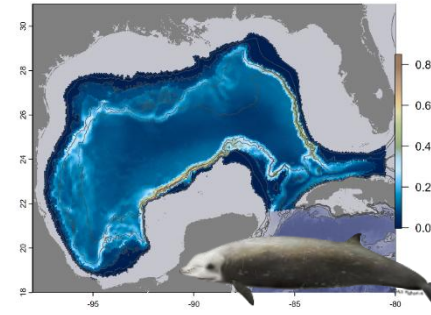
Abundance estimates for stock assessment

Gulf of Mexico — Atlantic spotted dolphin sightings (all years)
n = 83 sightings | Color: Winter (light) → Fall (dark)

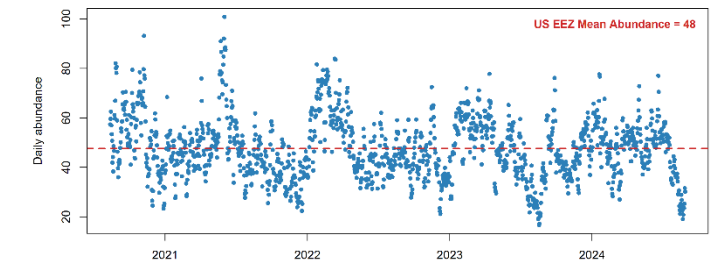
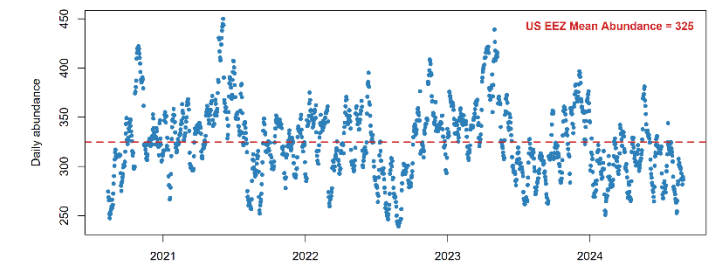
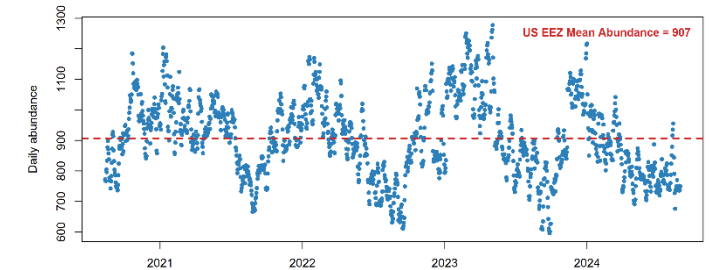


Integrating multi-modal data

NOAA ship-based visual surveys

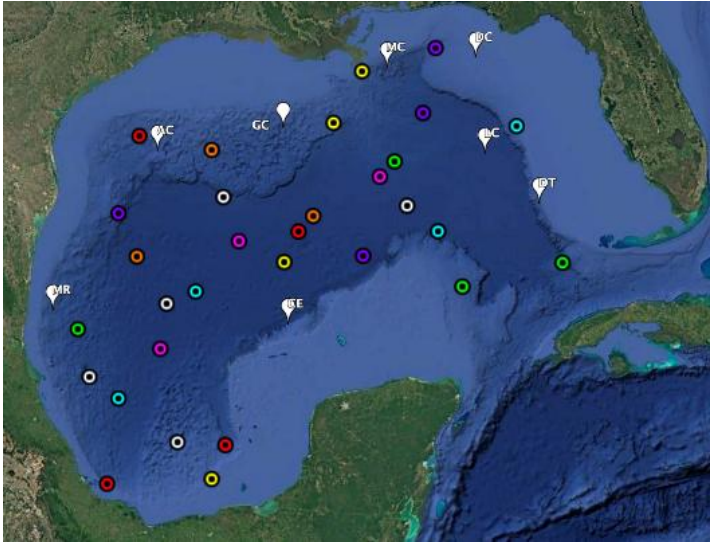


Preliminary estimates



Towards a comprehensive understanding of species distribution and abundance

PAM Monitoring from 2020-2029



Sustaining this observing capability would preserve the long-term record needed to track changing distributions and support future stock assessments

Next 5 years under LISTEN



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